Ecole Polytechnique–ESPCI 2026
Mathematics B
V2 Concours-Grade Correction

Detailed English correction

Contents

General conventions

All spectra below are counted with algebraic multiplicity, as in the statement. For a real symmetric matrix $M \in M_n(\mathbb{R})$ and a function f defined on the spectrum,

$$S_f(M) = \frac{1}{n} \sum_{(\lambda, m_\lambda) \in \mathrm{Sp}(M)} m_\lambda f(\lambda).$$

In Part IV we write $S_n(f) = S_f(X_n)$. The function f_k is always $x \mapsto x^k$.

This correction gives the details that are usually expected in a concours solution: endpoint cases, multiplicities, convergence estimates, and the probabilistic variance computations are written explicitly. The appendix records the corresponding Lean formalisation status; it is an audit aid, not a substitute for the mathematical correction.

1 Preliminary question

Question 1. The recurrence

$$u_0 = 0, \qquad u_1 = 1, \qquad u_n = \alpha u_{n-1} - u_{n-2} \quad (n \geq 2)$$

determines the sequence uniquely: if two sequences have the same first two terms and satisfy the same recurrence, they are equal by induction on n. Thus, in each case below, it is enough to exhibit a sequence with the announced formula and to check its first two values and the recurrence.

Case $|\alpha| > 2$. Put

$$r_+ = \frac{\alpha + \sqrt{\alpha^2 - 4}}{2}, \qquad r_- = \frac{\alpha - \sqrt{\alpha^2 - 4}}{2}.$$

Since $|\alpha| > 2$, we have $\alpha^2 - 4 > 0$, so $r_+ \neq r_-$ and $r_+ - r_- = \sqrt{\alpha^2 - 4}$. Moreover r_+ and r_- are the two roots of $r^2 - \alpha r + 1 = 0$, hence

$$r_\pm^{n+2} = \alpha r_\pm^{n+1} - r_\pm^n \qquad (n \geq 0).$$

Therefore the sequence

$$v_n = \frac{r_+^n - r_-^n}{r_+ - r_-}$$

satisfies the same recurrence. Also $v_0 = 0$ and $v_1 = 1$. By uniqueness,

$$\boxed{u_n = \frac{r_+^n - r_-^n}{r_+ - r_-} = \frac{r_+^n - r_-^n}{\sqrt{\alpha^2 - 4}} \qquad (n \geq 0).}$$

Case $|\alpha| < 2$. There is a unique $\theta \in (0, \pi)$ such that $\alpha = 2\cos\theta$. Here $\sin\theta > 0$. Define

$$v_n = \frac{\sin(n\theta)}{\sin\theta}.$$

Then $v_0 = 0$ and $v_1 = 1$. The addition formula gives, for every $n \geq 0$,

$$\sin((n+2)\theta) = 2\cos\theta\, \sin((n+1)\theta) - \sin(n\theta).$$

Dividing by $\sin\theta$ and using $\alpha = 2\cos\theta$ shows that $v_{n+2} = \alpha v_{n+1} - v_n$. Thus, by uniqueness,

$$\boxed{\begin{array}{c} u_n = \dfrac{\sin(n\theta)}{\sin\theta}, \\ \alpha = 2\cos\theta, \qquad 0 < \theta < \pi. \end{array}}$$

Case $\alpha = 2$**.** Let $v_n = n$. Then $v_0 = 0$, $v_1 = 1$, and

$$2v_{n+1} - v_n = 2(n+1) - n = n + 2 = v_{n+2}.$$

Hence

$$\boxed{u_n = n.}$$

Case $\alpha = -2$**.** Let $v_n = (-1)^{n+1} n$. Again $v_0 = 0$ and $v_1 = 1$. Moreover

$$-2v_{n+1} - v_n = -2(-1)^{n+2}(n+1) - (-1)^{n+1} n = (-1)^{n+3}(n+2) = v_{n+2}.$$

Therefore

$$\boxed{u_n = (-1)^{n+1} n.}$$

2 Part I

Question 2. All estimates below are written for $n \geq 1$; changing finitely many initial terms has no effect on the limit. Let f be continuous on $[0,1]$. Since $[0,1]$ is compact, f is uniformly continuous and bounded. Set

$$M = \sup_{[0,1]} |f|.$$

We shall use the following elementary Riemann-sum estimate, with the omitted right endpoint treated explicitly. Let $m \geq 1$, let $h > 0$ satisfy $mh \leq 1$, and define

$$R(m,h) = h \sum_{k=1}^{m} f(kh).$$

Then

$$\left| R(m,h) - \int_0^1 f(t)\,dt \right| \leq \sum_{k=1}^{m} \int_{(k-1)h}^{kh} |f(kh) - f(t)|\,dt + \int_{mh}^{1} |f(t)|\,dt.$$

For $t \in [(k-1)h, kh]$, both t and kh belong to $[0,1]$ and their distance is at most h. If

$$\omega(\delta) = \sup\{|f(x) - f(y)| : \ x, y \in [0,1], \ |x - y| \leq \delta\},$$

then $\omega(\delta) \to 0$ as $\delta \to 0$, and the previous bound gives

$$\left| R(m,h) - \int_0^1 f(t)\,dt \right| \leq mh\,\omega(h) + M(1 - mh).$$

The last term is exactly the contribution of the short interval $[mh, 1]$, whose length tends to 0 in the applications. Consequently, whenever $h_n \to 0$ and $m_n h_n \to 1$ with $m_n h_n \leq 1$, one has

$$h_n \sum_{k=1}^{m_n} f(kh_n) \longrightarrow \int_0^1 f(t)\,dt. \tag{2.1}$$

For the first sequence, take

$$h_n = \frac{1}{n+1}, \qquad m_n = n.$$

Then $h_n \to 0$, $m_n h_n = n/(n+1) \to 1$, and (2.1) yields

$$\frac{1}{n+1} \sum_{k=1}^{n} f\left(\frac{k}{n+1}\right) \longrightarrow \int_0^1 f(t)\,dt.$$

Since

$$v_n = \frac{n+1}{n}\left(\frac{1}{n+1}\sum_{k=1}^{n} f\left(\frac{k}{n+1}\right)\right)$$

and $(n+1)/n \to 1$, we get

$$v_n \longrightarrow \int_0^1 f(t)\,dt.$$

For the second sequence, take

$$h_n = \frac{2}{2n+1}, \qquad m_n = n.$$

Again $h_n \to 0$, $m_n h_n = 2n/(2n+1) \to 1$, and (2.1) gives

$$\frac{2}{2n+1}\sum_{k=1}^{n} f\left(\frac{2k}{2n+1}\right) \longrightarrow \int_0^1 f(t)\,dt.$$

But

$$w_n = \frac{2n+1}{2n}\left(\frac{2}{2n+1}\sum_{k=1}^{n} f\left(\frac{2k}{2n+1}\right)\right),$$

and $(2n+1)/(2n) \to 1$. Therefore

$$w_n \longrightarrow \int_0^1 f(t)\,dt.$$

Finally,

$$\boxed{\lim_{n\to\infty} v_n = \lim_{n\to\infty} w_n = \int_0^1 f(t)\,dt.}$$

Question 3a. Let f be continuous on $[-2,2]$ and let $M = \sup_{[-2,2]} |f|$. The only possible singularities of

$$\frac{f(x)}{\sqrt{4-x^2}}$$

are at $x=-2$ and $x=2$. Near 2, for $x \in [0,2)$,

$$4-x^2 = (2-x)(2+x) \geq 2(2-x),$$

so

$$\frac{|f(x)|}{\sqrt{4-x^2}} \leq \frac{M}{\sqrt{2}\sqrt{2-x}},$$

and $1/\sqrt{2-x}$ is integrable near 2. The endpoint -2 is treated similarly, using

$$4-x^2 = (2-x)(2+x) \geq 2(x+2) \qquad (x \in (-2,0]).$$

Thus the improper integral

$$I(f) = \frac{1}{\pi}\int_{-2}^{2} \frac{f(x)}{\sqrt{4-x^2}}\,dx$$

is convergent.

On $[-2+\varepsilon, 2-\varepsilon]$ we may use the substitution $x = 2\cos\theta$. Then

$$dx = -2\sin\theta\,d\theta, \qquad \sqrt{4-x^2} = 2\sin\theta \quad (0<\theta<\pi).$$

Letting $\varepsilon \downarrow 0$, justified by the convergence just proved, gives

$$\boxed{I(f) = \frac{1}{\pi}\int_0^{\pi} f(2\cos\theta)\,d\theta.}$$

Question 3b. For $f_0(x) = 1$,

$$I(f_0) = \frac{1}{\pi}\int_0^\pi 1\,d\theta = 1.$$

For $f_1(x) = x$,

$$I(f_1) = \frac{1}{\pi}\int_0^\pi 2\cos\theta\,d\theta = 0.$$

For $f_2(x) = x^2$,

$$I(f_2) = \frac{1}{\pi}\int_0^\pi 4\cos^2\theta\,d\theta = \frac{4}{\pi}\cdot\frac{\pi}{2} = 2.$$

Hence

$$\boxed{I(f_0) = 1, \qquad I(f_1) = 0, \qquad I(f_2) = 2.}$$

Question 3c. If n is odd, the function $x^n/\sqrt{4-x^2}$ is odd on $[-2,2]$, so

$$\boxed{I(f_n) = 0 \qquad (n \text{ odd}).}$$

Let $n = 2p$. Then

$$I(f_{2p}) = \frac{1}{\pi}\int_0^\pi (2\cos\theta)^{2p}\,d\theta.$$

It remains to compute $\int_0^\pi \cos^{2p}\theta\,d\theta$. Since $\cos^{2p}$ has period π,

$$\int_0^\pi \cos^{2p}\theta\,d\theta = \frac{1}{2}\int_0^{2\pi}\cos^{2p}\theta\,d\theta.$$

Using

$$\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2},$$

we get

$$\cos^{2p}\theta = 2^{-2p}\sum_{j=0}^{2p}\binom{2p}{j}e^{i(2p-2j)\theta}.$$

The integral over $[0,2\pi]$ kills every nonconstant exponential term. The constant term corresponds to $j = p$. Hence

$$\int_0^{2\pi}\cos^{2p}\theta\,d\theta = 2\pi\,2^{-2p}\binom{2p}{p},$$

and therefore

$$\int_0^{\pi}\cos^{2p}\theta\,d\theta = \pi\,2^{-2p}\binom{2p}{p}.$$

Consequently

$$I(f_{2p}) = \frac{2^{2p}}{\pi}\cdot\pi\,2^{-2p}\binom{2p}{p} = \binom{2p}{p}.$$

Thus

$$\boxed{I(f_n) = \begin{cases} 0, & n \text{ odd},\\ \dbinom{2p}{p}, & n = 2p.\end{cases}}$$

Question 4a. Let f be continuous on $[-2,2]$ and define

$$g(t) = f(2\cos(\pi t)), \qquad t \in [0,1].$$

Then g is continuous on $[0,1]$. Since U_n is uniform on $\{1,\dots,n\}$,

$$\begin{aligned}\mathbb{E}\left(f\left(2\cos\left(\frac{\pi U_n}{n+1}\right)\right)\right) &= \frac{1}{n}\sum_{k=1}^{n} f\left(2\cos\left(\frac{\pi k}{n+1}\right)\right)\\ &= \frac{1}{n}\sum_{k=1}^{n} g\left(\frac{k}{n+1}\right).\end{aligned}$$

By Question 2 this converges to

$$\int_0^1 g(t)\,dt = \frac{1}{\pi}\int_0^{\pi} f(2\cos\theta)\,d\theta = \frac{1}{\pi}\int_{-2}^{2}\frac{f(x)}{\sqrt{4-x^2}}\,dx.$$

Therefore

$$\boxed{\mathbb{E}\left(f\left(2\cos\left(\frac{\pi U_n}{n+1}\right)\right)\right) \longrightarrow I(f).}$$

Question 4b. Fix $y \in [-2,2]$ and define

$$\theta_y = \arccos\left(\frac{y}{2}\right) \in [0,\pi].$$

The function cos is strictly decreasing on $[0,\pi]$. Thus, for $1 \le k \le n$,

$$2\cos\left(\frac{\pi k}{n+1}\right) < y \iff \frac{\pi k}{n+1} > \theta_y.$$

If $y=-2$, then $\theta_y = \pi$ and the inequality is impossible, so the probability is 0 for every n. The right-hand integral is also 0. If $y = 2$, then $\theta_y = 0$ and the inequality is true for every $k = 1,\dots,n$, so the probability is 1 for every n. The right-hand integral is also 1.

Assume now $-2 < y < 2$. Put $a_n = (n+1)\theta_y/\pi$. The number of integers $k \in \{1,\dots,n\}$ with $k > a_n$ is $n - \lfloor a_n \rfloor$. Hence

$$\mathbb{P}\left(2\cos\left(\frac{\pi U_n}{n+1}\right) < y\right) = \frac{n-\lfloor a_n\rfloor}{n} \longrightarrow 1 - \frac{\theta_y}{\pi}.$$

Finally, with $x = 2\cos\theta$,

$$\frac{1}{\pi}\int_{-2}^{y}\frac{dx}{\sqrt{4-x^2}} = \frac{1}{\pi}\int_{\theta_y}^{\pi} d\theta = 1 - \frac{\theta_y}{\pi}.$$

Therefore, for every $y \in [-2,2]$,

$$\boxed{\mathbb{P}\left(2\cos\left(\frac{\pi U_n}{n+1}\right) < y\right) \longrightarrow \frac{1}{\pi}\int_{-2}^{y}\frac{dx}{\sqrt{4-x^2}}.}$$

3 Part II

Question 5a. For $n = 2$,

$$T_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \chi_2(X) = \det\begin{pmatrix} X & -1 \\ -1 & X \end{pmatrix} = X^2 - 1.$$

Thus

$$\boxed{\mathrm{Sp}(T_2) = \{-1, 1\}.}$$

For $n = 3$,

$$XI_3 - T_3 = \begin{pmatrix} X & -1 & 0 \\ -1 & X & -1 \\ 0 & -1 & X \end{pmatrix}.$$

Expanding the determinant gives

$$\chi_3(X) = X(X^2 - 1) - X = X^3 - 2X = X(X^2 - 2).$$

Hence

$$\boxed{\mathrm{Sp}(T_3) = \{-\sqrt{2}, 0, \sqrt{2}\}.}$$

Question 5b. It is convenient to set $\chi_0(X) = 1$ and $\chi_1(X) = X$. For $n \geq 2$, expand

$$\chi_n(X) = \det(XI_n - T_n)$$

along the first row. The first contribution is $X\chi_{n-1}(X)$. The only other nonzero entry in the first row is -1 in column 2; its cofactor has determinant $-\chi_{n-2}(X)$. Therefore

$$\boxed{\chi_n(X) = X\chi_{n-1}(X) - \chi_{n-2}(X) \qquad (n \geq 2).}$$

In particular this proves the requested recurrence for all $n \geq 4$.

Question 5c. Let $\alpha \in \mathbb{C}$ with $|\alpha| < 2$, and choose a square root Δ of $4 - \alpha^2$. Since $|\alpha| < 2$, one has $\alpha \neq \pm 2$, so $\Delta \neq 0$. Put

$$r_+ = \frac{\alpha + i\Delta}{2}, \qquad r_- = \frac{\alpha - i\Delta}{2}.$$

Then

$$r_+ + r_- = \alpha, \qquad r_+ r_- = \frac{\alpha^2 + \Delta^2}{4} = 1.$$

Thus r_+ and r_- are the two distinct roots of $r^2 - \alpha r + 1 = 0$.

For fixed α, the numbers $\chi_n(\alpha)$ satisfy

$$\chi_n(\alpha) = \alpha\chi_{n-1}(\alpha) - \chi_{n-2}(\alpha), \qquad \chi_0(\alpha) = 1, \qquad \chi_1(\alpha) = \alpha.$$

The sequence

$$q_n = \frac{r_+^{n+1} - r_-^{n+1}}{r_+ - r_-}$$

has the same two initial values and satisfies the same recurrence. Therefore $\chi_n(\alpha) = q_n$ for all $n \geq 0$. Since $r_+ - r_- = i\Delta$, we obtain, for every $n \geq 2$,

$$\boxed{\chi_n(\alpha) = \frac{1}{i\sqrt{4 - \alpha^2}} \left[\left(\frac{\alpha + i\sqrt{4 - \alpha^2}}{2} \right)^{n+1} - \left(\frac{\alpha - i\sqrt{4 - \alpha^2}}{2} \right)^{n+1} \right].}$$

Changing the sign of the chosen square root exchanges the two terms and changes the sign of the denominator, so the value is independent of this choice.

Question 5d. From the formula of Question 5c, expanded formally with $D^2 = 4 - X^2$, one obtains

$$\chi_n(X) = \frac{(X+iD)^{n+1} - (X-iD)^{n+1}}{iD\,2^{n+1}}$$
$$= \frac{1}{2^n} \sum_{s=0}^{\lfloor n/2 \rfloor} (-1)^s \binom{n+1}{2s+1} X^{n-2s}(4-X^2)^s.$$

Thus the coefficient of X^{n-2p}, for $0 \le p \le \lfloor n/2 \rfloor$, may be written as

$$\frac{(-1)^p 4^p}{2^n} \sum_{s=p}^{\lfloor n/2 \rfloor} \binom{n+1}{2s+1}\binom{s}{p}.$$

This is already an exact expression as a sum of products of binomial coefficients.

It simplifies to the following standard closed form:

$$\boxed{\chi_n(X) = \sum_{p=0}^{\lfloor n/2 \rfloor} (-1)^p \binom{n-p}{p} X^{n-2p}.}$$

For completeness, we verify the simplification by induction. The formula is true for $\chi_0 = 1$ and $\chi_1 = X$. If it holds for χ_{n-1} and χ_{n-2}, the coefficient of X^{n-2p} in

$$X\chi_{n-1}(X) - \chi_{n-2}(X)$$

is

$$(-1)^p \binom{n-1-p}{p} - (-1)^{p-1}\binom{n-p-1}{p-1} = (-1)^p \binom{n-p}{p},$$

by Pascal's identity. The induction follows.

Question 6. Let $\theta \in (0, \pi)$ and set $\alpha = 2\cos\theta$. By Question 1 or by Question 5c,

$$\chi_n(2\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}.$$

For

$$\theta_k = \frac{k\pi}{n+1}, \qquad k = 1, \dots, n,$$

we have $\sin\theta_k \neq 0$ and $\sin((n+1)\theta_k) = \sin(k\pi) = 0$. Therefore

$$2\cos\left(\frac{k\pi}{n+1}\right)$$

is a root of χ_n. These n numbers are distinct because cos is strictly decreasing on $[0, \pi]$. Since χ_n has degree n, they are all the roots. Hence

$$\boxed{\mathrm{Sp}(T_n) = \left\{2\cos\left(\frac{k\pi}{n+1}\right) : 1 \le k \le n\right\},}$$

each eigenvalue having multiplicity 1.

Question 7. By Question 6,

$$S_f(T_n) = \frac{1}{n}\sum_{k=1}^{n} f\left(2\cos\left(\frac{k\pi}{n+1}\right)\right).$$

This is exactly the expectation in Question 4a with U_n uniform on $\{1, \dots, n\}$. Therefore, for every continuous f on $[-2, 2]$,

$$\boxed{\lim_{n\to\infty} S_f(T_n) = \frac{1}{\pi}\int_{-2}^{2} \frac{f(x)}{\sqrt{4-x^2}}\,dx.}$$

Question 8a. We have

$$T_n(a,b,c) = aI_n + T_n(0,b,c).$$

If μ is an eigenvalue of $T_n(0,b,c)$, then $a+\mu$ is an eigenvalue of $T_n(a,b,c)$ with the same algebraic multiplicity. Equivalently,

$$\chi_{T_n(a,b,c)}(X) = \chi_{T_n(0,b,c)}(X-a).$$

Thus

$$\boxed{\mathrm{Sp}(T_n(a,b,c)) = a + \mathrm{Sp}(T_n(0,b,c)),}$$

with multiplicities preserved.

Question 8b. Let

$$D_n(X) = \det(XI_n - T_n(0,b,c)).$$

The matrix $XI_n - T_n(0,b,c)$ has diagonal entries X, superdiagonal entries $-b$, and subdiagonal entries $-c$. Expanding as in Question 5b gives

$$D_n(X) = XD_{n-1}(X) - bc\, D_{n-2}(X), \qquad D_0 = 1, \qquad D_1 = X.$$

The same recurrence and the same initial values are obtained for

$$\det(XI_n - T_n(0,bc,1)).$$

Therefore the two characteristic polynomials are equal. Together with Question 8a, this gives

$$\boxed{\mathrm{Sp}(T_n(a,b,c)) = a + \mathrm{Sp}(T_n(0,bc,1)),}$$

again with algebraic multiplicities.

Question 8c. Assume $bc > 0$. Then b and c have the same sign. Set

$$r = \sqrt{\frac{c}{b}} > 0, \qquad D = \mathrm{diag}(1, r, r^2, \dots, r^{n-1}).$$

A direct computation gives

$$D^{-1}T_n(0,b,c)D = s\sqrt{bc}\, T_n,$$

where $s = 1$ if $b, c > 0$ and $s = -1$ if $b, c < 0$. Indeed the new superdiagonal and subdiagonal entries are respectively br and c/r, both equal to $s\sqrt{bc}$.

Thus $T_n(a,b,c)$ is similar to

$$aI_n + s\sqrt{bc}\, T_n.$$

By Question 6 its eigenvalues are

$$a + 2s\sqrt{bc}\cos\left(\frac{k\pi}{n+1}\right), \qquad k = 1, \dots, n.$$

If $s = -1$, this list is the same as the list with $s = 1$, because replacing k by $n+1-k$ changes the sign of the cosine. Hence, in all cases $bc > 0$,

$$\boxed{\lambda_k = a + 2\sqrt{bc}\cos\left(\frac{k\pi}{n+1}\right), \qquad k = 1, \dots, n.}$$

The eigenvalues are real and simple.

Question 9a. For $bc > 0$, Question 8c gives

$$S_f(T_n(a,b,c)) = \frac{1}{n}\sum_{k=1}^{n} f\left(a + 2\sqrt{bc}\cos\left(\frac{k\pi}{n+1}\right)\right).$$

Define g on $[-2,2]$ by

$$g(x) = f(a + \sqrt{bc}\,x).$$

If f is continuous on $\mathbb{R}$, then g is continuous on $[-2,2]$. Applying Question 7 to g gives

$$\begin{aligned}\lim_{n\to\infty} S_f(T_n(a,b,c)) &= \frac{1}{\pi}\int_{-2}^{2} \frac{g(x)}{\sqrt{4-x^2}}\,dx \\ &= \frac{1}{\pi}\int_{-2}^{2} \frac{f(a+\sqrt{bc}\,x)}{\sqrt{4-x^2}}\,dx.\end{aligned}$$

Thus

$$\boxed{\lim_{n\to\infty} S_f(T_n(a,b,c)) = \frac{1}{\pi}\int_{-2}^{2} \frac{f(a+\sqrt{bc}\,x)}{\sqrt{4-x^2}}\,dx.}$$

Question 9b. Let

$$\lambda_{n,k} = a + 2\sqrt{bc}\cos\left(\frac{k\pi}{n+1}\right), \qquad 1 \le k \le n.$$

The sequence $k \mapsto \lambda_{n,k}$ is strictly decreasing. Define

$$L_- = a - 2\sqrt{bc}, \qquad L_+ = a + 2\sqrt{bc}.$$

If $y < L_-$, then $q_n(y) = 0$ for every n. If $y \ge L_+$, then $q_n(y) = n$ for every n.
Assume now $L_- < y < L_+$. Put

$$\theta_y = \arccos\left(\frac{y-a}{2\sqrt{bc}}\right) \in (0,\pi).$$

Then

$$\lambda_{n,k} \le y \iff \frac{k\pi}{n+1} \ge \theta_y.$$

Hence

$$q_n(y) = \#\left\{1 \le k \le n : k \ge \frac{n+1}{\pi}\theta_y\right\} = n - \left\lceil \frac{n+1}{\pi}\theta_y \right\rceil + 1,$$

up to the harmless convention when the ceiling equals $n+1$, which cannot occur in the present open interval for large n. Dividing by n gives

$$\frac{q_n(y)}{n} \longrightarrow 1 - \frac{\theta_y}{\pi}.$$

Equivalently,

$$1 - \frac{\theta_y}{\pi} = \frac{1}{\pi}\int_{-2}^{(y-a)/\sqrt{bc}} \frac{dx}{\sqrt{4-x^2}}.$$

Therefore the clean global statement is

$$\boxed{\frac{q_n(y)}{n} \longrightarrow F(y),}$$

where

$$F(y) = \begin{cases} 0, & y \le a - 2\sqrt{bc}, \\ \dfrac{1}{\pi}\displaystyle\int_{-2}^{(y-a)/\sqrt{bc}} \frac{dx}{\sqrt{4-x^2}}, & a - 2\sqrt{bc} < y < a + 2\sqrt{bc}, \\ 1, & y \ge a + 2\sqrt{bc}. \end{cases}$$

Consequently, whenever $F(y) > 0$,

$$\boxed{q_n(y) \sim nF(y).}$$

At the lower edge $y \le a - 2\sqrt{bc}$ the count is identically zero, so one should not state a nonzero equivalent there.

4 Part III: constructive Weierstrass approximation

The Weierstrass approximation theorem is not used in this part. It is proved from the polynomials

$$Q_n(X) = (1 - X^n)^{2^n}, \qquad P_n(X) = Q_n\left(\frac{1-X}{2}\right).$$

Question 10a. Let $0 \le \kappa < 1/2$.

On $[0, \kappa]$, for $n \ge 1$ and $0 \le x \le \kappa$,

$$0 \le x^n \le \kappa^n.$$

Using $1 - (1-t)^m \le mt$ for $0 \le t \le 1$ and $m \ge 1$, we obtain

$$0 \le 1 - Q_n(x) = 1 - (1 - x^n)^{2^n} \le 2^n x^n \le (2\kappa)^n.$$

Since $2\kappa < 1$, this tends uniformly to 0. Thus

$$Q_n \longrightarrow 1 \quad \text{uniformly on } [0, \kappa].$$

On $[1-\kappa, 1]$, we have $x^n \ge (1-\kappa)^n$. Hence

$$0 \le Q_n(x) = (1 - x^n)^{2^n} \le \left(1 - (1-\kappa)^n\right)^{2^n}.$$

Using $1 - t \le e^{-t}$,

$$Q_n(x) \le \exp\left(-2^n (1-\kappa)^n\right) = \exp\left(-(2(1-\kappa))^n\right).$$

Since $2(1-\kappa) > 1$, this upper bound tends to 0. Thus

$$Q_n \longrightarrow 0 \quad \text{uniformly on } [1-\kappa, 1].$$

Question 10b. Let $0 < \eta \le 1$. For $x \in [-1, 1]$ put

$$t = \frac{1-x}{2}.$$

Then $P_n(x) = Q_n(t)$.

If $x \in [\eta, 1]$, then

$$0 \le t \le \frac{1-\eta}{2} =: \kappa < \frac{1}{2},$$

so $Q_n(t) \to 1$ uniformly by Question 10a. Since $H(x) = 1$ on $[\eta, 1]$, this gives uniform convergence to H there.

If $x \in [-1, -\eta]$, then

$$\frac{1+\eta}{2} \le t \le 1.$$

Writing $\kappa = (1-\eta)/2 < 1/2$, this interval is contained in $[1-\kappa, 1]$, so $Q_n(t) \to 0$ uniformly by Question 10a. Since $H(x) = 0$ on $[-1, -\eta]$, this gives uniform convergence to H there.

Therefore

$$\boxed{P_n \longrightarrow H \quad \text{uniformly on } [-1,1] \setminus [-\eta, \eta].}$$

Question 11. Assume first that $f(-1) = 0$. Since f is continuous on the compact interval $[-1, 1]$, it is uniformly continuous. Choose $\delta > 0$ such that

$$|x - y| \le \delta \quad \Longrightarrow \quad |f(x) - f(y)| \le \varepsilon.$$

Choose a partition

$$-1 = t_0 < t_1 < \cdots < t_N < t_{N+1} = 1$$

with mesh at most δ, and set

$$c_i = t_i, \qquad a_i = f(t_i) - f(t_{i-1}) \qquad (1 \le i \le N).$$

Then $-1 < c_1 < \cdots < c_N < 1$, and

$$|a_i| = |f(t_i) - f(t_{i-1})| \le \varepsilon.$$

Define

$$S(x) = \sum_{i=1}^{N} a_i H(x - c_i).$$

If $x \in [t_j, t_{j+1})$ with $0 \le j \le N$, then exactly the jumps $c_1, \ldots, c_j$ have occurred, so

$$S(x) = \sum_{i=1}^{j} a_i = f(t_j) - f(t_0) = f(t_j),$$

because $f(t_0) = f(-1) = 0$. Hence

$$|f(x) - S(x)| = |f(x) - f(t_j)| \le \varepsilon.$$

At $x = 1$, one has $S(1) = f(t_N)$ and therefore

$$|f(1) - S(1)| = |f(1) - f(t_N)| \le \varepsilon.$$

Thus

$$\boxed{\forall x \in [-1,1], \qquad \left| f(x) - \sum_{i=1}^{N} a_i H(x - c_i) \right| \le \varepsilon,}$$

with $a_i \in [-\varepsilon, \varepsilon]$.

Question 12. We first prove the result for $f(-1) = 0$, then remove this assumption.

Let $\varepsilon > 0$ and assume $f(-1) = 0$. Apply Question 11 with tolerance $\varepsilon/4$. We obtain points

$$-1 < c_1 < \cdots < c_N < 1$$

and coefficients a_i with $|a_i| \le \varepsilon/4$ such that

$$|f(x) - S(x)| \le \frac{\varepsilon}{4}, \qquad S(x) = \sum_{i=1}^{N} a_i H(x - c_i).$$

For each i, choose a positive number ρ_i such that

$$\frac{x - c_i}{\rho_i} \in [-1, 1] \qquad \text{for all } x \in [-1, 1].$$

For example, one may take

$$\rho_i = 1 + |c_i|.$$

Then

$$H\left(\frac{x - c_i}{\rho_i}\right) = H(x - c_i),$$

because $\rho_i > 0$.

Choose $\eta > 0$ small enough that the intervals

$$[c_i - \rho_i \eta, c_i + \rho_i \eta]$$

are pairwise disjoint. This is possible because the points c_i are distinct. By Question 10b, for each fixed i,

$$P_m\left(\frac{x - c_i}{\rho_i}\right) \longrightarrow H\left(\frac{x - c_i}{\rho_i}\right) = H(x - c_i)$$

uniformly on the set of $x \in [-1, 1]$ such that $|x - c_i| \geq \rho_i \eta$.

Let

$$A = \sum_{i=1}^{N} |a_i|.$$

If $A = 0$, then $S = 0$ and the proof is immediate. Otherwise choose m large enough that, simultaneously for all i and all $x \in [-1, 1]$ with $|x - c_i| \geq \rho_i \eta$,

$$\left| P_m\left(\frac{x - c_i}{\rho_i}\right) - H(x - c_i) \right| \leq \frac{\varepsilon}{4A}.$$

Define the polynomial

$$R_m(X) = \sum_{i=1}^{N} a_i P_m\left(\frac{X - c_i}{\rho_i}\right).$$

Fix $x \in [-1, 1]$. Because the intervals $[c_i - \rho_i \eta, c_i + \rho_i \eta]$ are pairwise disjoint, x belongs to at most one of them. For all indices for which $|x - c_i| \geq \rho_i \eta$, the total error is bounded by

$$\sum_i |a_i| \frac{\varepsilon}{4A} \leq \frac{\varepsilon}{4}.$$

For the possible exceptional index j, we use only the elementary bound

$$0 \leq P_m(t) \leq 1 \qquad (-1 \leq t \leq 1),$$

which follows from $0 \leq (1 - t)/2 \leq 1$. Hence

$$\left| P_m\left(\frac{x - c_j}{\rho_j}\right) - H(x - c_j) \right| \leq 1,$$

and its contribution is at most $|a_j| \leq \varepsilon/4$. Thus

$$|R_m(x) - S(x)| \leq \frac{\varepsilon}{2}.$$

Together with $|f(x) - S(x)| \leq \varepsilon/4$, this gives

$$|f(x) - R_m(x)| \leq \frac{3\varepsilon}{4} < \varepsilon.$$

This proves the result when $f(-1) = 0$.

For a general continuous f, apply the previous case to

$$g(x) = f(x) - f(-1),$$

which satisfies $g(-1) = 0$. There is a polynomial R such that

$$|g(x) - R(x)| \leq \varepsilon \qquad (x \in [-1, 1]).$$

Then

$$P(X) = R(X) + f(-1)$$

satisfies

$$\boxed{\forall x \in [-1, 1], \qquad |f(x) - P(x)| \leq \varepsilon.}$$

Scaling note. The proof above uses the scaled polynomials $P_m((X - c_i)/\rho_i)$. This is the same construction as the indicated one, but it keeps the argument of P_m inside $[-1, 1]$, where Question 10b applies. Since $\rho_i > 0$, the step function is unchanged: $H((x - c_i)/\rho_i) = H(x - c_i)$.

5 Part IV

For $k \geq 0$, recall

$$f_k(x) = x^k, \qquad \Sigma(f) = \frac{1}{2\pi} \int_{-2}^{2} f(x) \sqrt{4 - x^2}\, dx.$$

For later use, if $c \in \mathbb{R}$ and $j \geq 0$, the function

$$x \longmapsto c\, x^j \sqrt{4 - x^2}$$

is continuous on the compact interval $[-2, 2]$, hence integrable there. Thus finite linear combinations of monomials may be integrated term by term against the semicircle weight.

Question 13a. For every ω, the matrix $X_n(\omega)$ is real symmetric. Hence it is diagonalizable over $\mathbb{R}$ in an orthonormal basis. If its eigenvalues are $\lambda_1, \ldots, \lambda_n$, repeated with multiplicity, then the eigenvalues of $X_n(\omega)^k$ are $\lambda_1^k, \ldots, \lambda_n^k$. Therefore

$$\mathrm{Tr}(X_n(\omega)^k) = \sum_{j=1}^{n} \lambda_j^k.$$

On the other hand,

$$S_n(f_k)(\omega) = \frac{1}{n} \sum_{j=1}^{n} \lambda_j^k.$$

Thus, as random variables,

$$\boxed{S_n(f_k) = \frac{1}{n} \mathrm{Tr}(X_n^k).}$$

Question 13b. If k is odd, the integrand $x^k \sqrt{4 - x^2}$ is odd on $[-2, 2]$, so

$$\boxed{\Sigma(f_k) = 0 \qquad (k \text{ odd}).}$$

Let $k = 2p$. With $x = 2\cos\theta$,

$$dx = -2 \sin\theta\, d\theta, \qquad \sqrt{4 - x^2} = 2 \sin\theta,$$

and therefore

$$\begin{aligned}\Sigma(f_{2p}) &= \frac{1}{2\pi}\int_0^\pi (2\cos\theta)^{2p}\, 4\sin^2\theta\, d\theta \\ &= \frac{2^{2p+1}}{\pi}\int_0^\pi \cos^{2p}\theta\, \sin^2\theta\, d\theta.\end{aligned}$$

From Question 3c,

$$A_p := \int_0^\pi \cos^{2p}\theta\, d\theta = \pi\, 4^{-p}\binom{2p}{p}.$$

Since

$$\int_0^\pi \cos^{2p}\theta\,\sin^2\theta\, d\theta = A_p - A_{p+1},$$

and

$$\frac{A_{p+1}}{A_p} = \frac{2p+1}{2p+2},$$

we obtain

$$A_p - A_{p+1} = \frac{A_p}{2p+2}.$$

Thus

$$\begin{aligned}\Sigma(f_{2p}) &= \frac{2^{2p+1}}{\pi}\cdot\frac{1}{2p+2}\pi\, 4^{-p}\binom{2p}{p} \\ &= \frac{1}{p+1}\binom{2p}{p}.\end{aligned}$$

Consequently

$$\boxed{\Sigma(f_k) = \begin{cases} 0, & k \text{ odd}, \\ \dfrac{1}{p+1}\dbinom{2p}{p}, & k = 2p.\end{cases}}$$

The even moments are the Catalan numbers.

Question 13c. We prove (H_k) for $k = 0, 1, 2$.

Case $k = 0$. Since $X_n^0 = I_n$,

$$\mathrm{Tr}(X_n^0) = n.$$

Therefore

$$\frac{1}{n}\mathbb{E}(\mathrm{Tr}(X_n^0)) = 1 = \Sigma(f_0),$$

and

$$\frac{1}{n^2}\mathbb{E}(\mathrm{Tr}(X_n^0)^2) = \frac{1}{n^2}n^2 = 1 = \Sigma(f_0)^2.$$

Thus (H_0) holds.

Case $k = 1$. We have

$$\mathrm{Tr}(X_n) = \frac{1}{\sqrt{n}}\sum_{i=1}^n W_{i,i}.$$

Since the variables have mean zero,

$$\frac{1}{n}\mathbb{E}(\mathrm{Tr}(X_n)) = 0 = \Sigma(f_1).$$

Furthermore, by independence and mean zero, the cross terms vanish:

$$\begin{aligned}\mathbb{E}(\mathrm{Tr}(X_n)^2) &= \frac{1}{n}\mathbb{E}\left(\sum_{i=1}^{n} W_{i,i}\right)^2 \\ &= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(W_{i,i}^2) = \frac{1}{n}\cdot n = 1.\end{aligned}$$

Hence

$$\frac{1}{n^2}\mathbb{E}(\mathrm{Tr}(X_n)^2) = \frac{1}{n^2} \longrightarrow 0 = \Sigma(f_1)^2.$$

Thus (H_1) holds.

Case $k = 2$. Since X_n is symmetric,

$$\mathrm{Tr}(X_n^2) = \sum_{i,j=1}^{n} (X_n)_{i,j}^2.$$

Using the definition of X_n,

$$\mathrm{Tr}(X_n^2) = \frac{1}{n}\left(\sum_{i=1}^{n} W_{i,i}^2 + 2\sum_{1\le i<j\le n} W_{i,j}^2\right).$$

Thus

$$\begin{aligned}\mathbb{E}(\mathrm{Tr}(X_n^2)) &= \frac{1}{n}\left(n\mathbb{E}(W_{1,1}^2) + 2\binom{n}{2}\mathbb{E}(W_{1,1}^2)\right) \\ &= \frac{1}{n}\left(n + n(n-1)\right) = n.\end{aligned}$$

Therefore

$$\frac{1}{n}\mathbb{E}(\mathrm{Tr}(X_n^2)) = 1 = \Sigma(f_2).$$

It remains to prove the second limit. Put

$$Y_{i,j} = W_{i,j}^2, \qquad A_n = \sum_{i=1}^{n} Y_{i,i} + 2\sum_{1\le i<j\le n} Y_{i,j}.$$

Then

$$\mathrm{Tr}(X_n^2) = \frac{A_n}{n}, \qquad \mathbb{E}(A_n) = n^2.$$

Since $|W_{1,1}|^4$ has finite expectation, $Y_{1,1}$ has finite variance. Let

$$\sigma_Y^2 = \mathrm{Var}(Y_{1,1}) < \infty.$$

By independence of the variables $W_{i,j}$ for $i \le j$,

$$\mathrm{Var}(A_n) = n\sigma_Y^2 + 4\binom{n}{2}\sigma_Y^2 = O(n^2).$$

Consequently

$$\mathbb{E}(A_n^2) = \mathbb{E}(A_n)^2 + \mathrm{Var}(A_n) = n^4 + O(n^2).$$

Hence

$$\frac{1}{n^2}\mathbb{E}(\mathrm{Tr}(X_n^2)^2) = \frac{1}{n^2}\mathbb{E}\left(\frac{A_n^2}{n^2}\right) = \frac{\mathbb{E}(A_n^2)}{n^4} \longrightarrow 1 = \Sigma(f_2)^2.$$

Thus (H_2) holds.

In the remainder of the part, as stated in the problem, we assume (H_k) for every $k \ge 0$.

Question 14. Let $k \geq 0$, $B > 0$, and

$$g_{k,B}(x) = |x|^k \mathbf{1}_{|x|>B}.$$

For every real x,

$$|x|^k \mathbf{1}_{|x|>B} \leq \frac{|x|^{2k}}{B^k}.$$

Indeed, on $\{|x| > B\}$ this is equivalent to $B^k \leq |x|^k$, and outside this set the left-hand side is zero. Therefore, for every ω,

$$S_n(g_{k,B})(\omega) \leq \frac{1}{B^k} S_n(f_{2k})(\omega).$$

The random variable $S_n(g_{k,B})$ is nonnegative. Markov's inequality gives

$$\begin{aligned}\mathbb{P}(S_n(g_{k,B}) \geq \varepsilon) &\leq \frac{1}{\varepsilon}\mathbb{E}(S_n(g_{k,B})) \\ &\leq \frac{1}{\varepsilon B^k}\mathbb{E}(S_n(f_{2k})).\end{aligned}$$

Hence

$$\boxed{\mathbb{P}(S_n(g_{k,B}) \geq \varepsilon) \leq \frac{\mathbb{E}(S_n(f_{2k}))}{\varepsilon B^k}.}$$

Question 15. Fix $k \geq 0$, $B > 4$, and $\varepsilon > 0$. Let $\ell \geq k$. Since $|x| > B > 1$ implies $|x|^k \leq |x|^\ell$, we have

$$g_{k,B} \leq g_{\ell,B}.$$

Therefore

$$\mathbb{P}(S_n(g_{k,B}) \geq \varepsilon) \leq \mathbb{P}(S_n(g_{\ell,B}) \geq \varepsilon).$$

By Question 14,

$$\mathbb{P}(S_n(g_{\ell,B}) \geq \varepsilon) \leq \frac{\mathbb{E}(S_n(f_{2\ell}))}{\varepsilon B^\ell}.$$

Using Question 13a and hypothesis $(H_{2\ell})$,

$$\mathbb{E}(S_n(f_{2\ell})) \longrightarrow \Sigma(f_{2\ell}).$$

By Question 13b,

$$\Sigma(f_{2\ell}) = \frac{1}{\ell+1}\binom{2\ell}{\ell} \leq \binom{2\ell}{\ell} \leq 4^\ell.$$

Taking lim sup in n gives

$$\limsup_{n\to\infty} \mathbb{P}(S_n(g_{k,B}) \geq \varepsilon) \leq \frac{4^\ell}{\varepsilon B^\ell} = \frac{1}{\varepsilon}\left(\frac{4}{B}\right)^\ell.$$

Now let $\ell \to \infty$. Since $B > 4$, the right-hand side tends to 0. Hence

$$\boxed{\mathbb{P}(S_n(g_{k,B}) \geq \varepsilon) \longrightarrow 0.}$$

Question 16. Let

$$Z_n = S_n(f_k) = \frac{1}{n}\operatorname{Tr}(X_n^k).$$

By (H_k),

$$\mathbb{E}(Z_n) = \frac{1}{n}\mathbb{E}(\operatorname{Tr}(X_n^k)) \longrightarrow \Sigma(f_k),$$

and

$$\mathbb{E}(Z_n^2) = \frac{1}{n^2}\mathbb{E}(\operatorname{Tr}(X_n^k)^2) \longrightarrow \Sigma(f_k)^2.$$

Therefore

$$\operatorname{Var}(Z_n) = \mathbb{E}(Z_n^2) - \mathbb{E}(Z_n)^2 \longrightarrow 0.$$

Chebyshev's inequality gives

$$\mathbb{P}(|Z_n - \mathbb{E}Z_n| \geq \varepsilon) \leq \frac{\operatorname{Var}(Z_n)}{\varepsilon^2} \longrightarrow 0.$$

Thus

$$\boxed{\mathbb{P}(|S_n(f_k) - \mathbb{E}(S_n(f_k))| \geq \varepsilon) \longrightarrow 0.}$$

Question 17a. Fix $B > 4$, let f be continuous on $\mathbb{R}$ and zero outside $[-B, B]$, and fix $\varepsilon > 0$.

We shall use explicitly the following elementary properties of the spectral average. For each fixed matrix, S_n is a positive linear functional on functions of the eigenvalue variable:

$$S_n(af + bg) = aS_n(f) + bS_n(g), \qquad f \leq g \Rightarrow S_n(f) \leq S_n(g),$$

and it is normalised by $S_n(1) = 1$. Consequently, if $|h| \leq g$, then

$$|S_n(h)| \leq S_n(g).$$

These identities follow immediately from the definition $S_n(h) = n^{-1}\sum_{i=1}^n h(\lambda_i)$, with eigenvalues repeated according to multiplicity.

Apply Question 12 to the function $t \mapsto f(Bt)$ on $[-1, 1]$, then compose the approximating polynomial with $x \mapsto x/B$. Thus there is a polynomial $P \in \mathbb{R}[X]$ such that

$$\sup_{|x| \leq B} |f(x) - P(x)| \leq \eta,$$

where $\eta > 0$ will be chosen below. We take

$$\eta = \frac{\varepsilon}{16}.$$

Since the semicircle density is supported on $[-2, 2] \subset [-B, B]$, we have

$$|\Sigma(f) - \Sigma(P)| \leq \eta\Sigma(1) = \eta.$$

Indeed, on $[-2, 2]$ one has $|f - P| \leq \eta$ and $\sqrt{4 - x^2} \geq 0$, hence

$$|\Sigma(f) - \Sigma(P)| \leq \frac{1}{2\pi}\int_{-2}^{2} |f(x) - P(x)|\sqrt{4 - x^2}\, dx \leq \eta\Sigma(1).$$

Here $\Sigma(1) = 1$ by Question 13b with $p = 0$. Moreover, because P is a finite linear combination of monomials, linearity of S_n, linearity of expectation, and the first part of (H_k) give

$$\mathbb{E}(S_n(P)) \longrightarrow \Sigma(P).$$

Choose N such that, for all $n \geq N$,

$$|\mathbb{E}(S_n(P)) - \Sigma(P)| \leq \eta.$$

For every real x,

$$f(x) - P(x) = (f(x) - P(x))\mathbf{1}_{|x|\leq B} - P(x)\mathbf{1}_{|x|>B},$$

because $f(x) = 0$ when $|x| > B$. Therefore, for every n and every ω,

$$|S_n(f)(\omega) - S_n(P)(\omega)| \leq \eta + |S_n(P\mathbf{1}_{|x|>B})(\omega)|.$$

Indeed, the part $(f - P)\mathbf{1}_{|x|\leq B}$ has absolute value at most η, so positivity and $S_n(1) = 1$ bound its spectral average by η; the remaining term is the polynomial tail. For $n \geq N$, we now write

$$\begin{aligned}|S_n(f) - \Sigma(f)| &\leq |S_n(P) - \mathbb{E}(S_n(P))| \\ &\quad + |\mathbb{E}(S_n(P)) - \Sigma(P)| + |\Sigma(P) - \Sigma(f)| + |S_n(f) - S_n(P)| \\ &\leq |S_n(P) - \mathbb{E}(S_n(P))| + |S_n(P\mathbf{1}_{|x|>B})| + 3\eta.\end{aligned}$$

Since $3\eta = 3\varepsilon/16 < \varepsilon/2$, if both

$$|S_n(P\mathbf{1}_{|x|>B})| < \frac{\varepsilon}{4} \quad \text{and} \quad |S_n(P) - \mathbb{E}(S_n(P))| < \frac{\varepsilon}{4},$$

then

$$|S_n(f) - \Sigma(f)| < \varepsilon.$$

Consequently, for every $n \geq N$,

$$\boxed{\mathbb{P}(|S_n(f) - \Sigma(f)| \geq \varepsilon) \leq \mathbb{P}(|S_n(P\mathbf{1}_{|x|>B})| \geq \varepsilon/4) + \mathbb{P}(|S_n(P) - \mathbb{E}(S_n(P))| \geq \varepsilon/4).}$$

Question 17b. We prove that the two probabilities on the right-hand side of Question 17a tend to zero.

First, write

$$P(x) = \sum_{j=0}^{d} c_j x^j.$$

By Question 16 and the union bound, polynomial fluctuations vanish in probability. Indeed, if

$$C_1 = \sum_{j=0}^{d} |c_j|,$$

then

$$|S_n(P) - \mathbb{E}(S_n(P))| \leq \sum_{j=0}^{d} |c_j|\,|S_n(f_j) - \mathbb{E}(S_n(f_j))|.$$

If $C_1 = 0$, the quantity is zero. Otherwise, the event

$$|S_n(P) - \mathbb{E}(S_n(P))| \geq \varepsilon/4$$

is contained in the union of the events

$$|S_n(f_j) - \mathbb{E}(S_n(f_j))| \geq \frac{\varepsilon}{4(d+1)\max(1, C_1)} \qquad (0 \leq j \leq d),$$

each of which has probability tending to zero by Question 16. Hence

$$\mathbb{P}(|S_n(P) - \mathbb{E}(S_n(P))| \geq \varepsilon/4) \longrightarrow 0.$$

Second, for $|x| > B > 1$ and $0 \le j \le d$, we have $|x|^j \le |x|^d$. Therefore

$$|P(x)|\mathbf{1}_{|x|>B} \le C_1 |x|^d \mathbf{1}_{|x|>B} = C_1 g_{d,B}(x).$$

Thus

$$|S_n(P\mathbf{1}_{|x|>B})| \le S_n(|P|\mathbf{1}_{|x|>B}) \le C_1 S_n(g_{d,B}).$$

The first inequality is the estimate $|S_n(h)| \le S_n(|h|)$ applied to $h = P\mathbf{1}_{|x|>B}$, and the second follows from positivity of S_n. If $C_1 = 0$, the probability is zero. Otherwise Question 15 gives

$$\begin{aligned}\mathbb{P}(|S_n(P\mathbf{1}_{|x|>B})| \ge \varepsilon/4) &\le \mathbb{P}\left(S_n(g_{d,B}) \ge \frac{\varepsilon}{4C_1}\right) \\ &\longrightarrow 0.\end{aligned}$$

Combining this with Question 17a yields

$$\boxed{\mathbb{P}(|S_n(f) - \Sigma(f)| \ge \varepsilon) \longrightarrow 0.}$$

This is the asserted convergence in probability of the spectral averages against every continuous compactly supported test function supported in $[-B, B]$ with $B > 4$.

A Lean formalisation notes

This appendix records the state of the Lean formalisation that accompanies the correction. It is deliberately separated from the main text: a concours correction should read as mathematics, whereas the Lean repository records which statements have been mechanised, which hypotheses remain explicit, and which proof routes were chosen for formal reasons.

Build status. The canonical Lean repository builds with

```
lake build.
```

The source tree `XENS2026MB` contains no `sorry`, `admit`, project `axiom`, or `unsafe` declaration. The audited endpoints checked during the last pass depend only on Lean's standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound`. This means there is no hidden project-specific axiom in the audited proof graph.

Ticket status. Of the 38 development tickets used for the formalisation, 37 are marked *done*. The completed tickets include the recurrence, the Riemann-sum and arcsine calculations, the constructive Weierstrass proof from the explicit polynomials Q_n and P_n, the path-matrix spectrum, the Toeplitz factorisation for $bc > 0$, the semicircle moments, the tail estimates, the polynomial concentration reduction, and the compact-support convergence argument.

The former low-degree model gap is now closed for the concrete Mathlib-usable Wigner model `WignerInputConcrete`, with measurable integer entries, all entry moments, centred upper entries, unit second moments, independent upper entries, identical distribution, and a uniform fourth-moment bound.

- **T31**: the low-degree moment hypotheses (H_1) and (H_2) are now supplied by `WignerInputConcrete.lowDeg`
  The last nontrivial field was the second degree-two trace moment

$$\mathbb{E}\left[\left(\frac{1}{n}\operatorname{Tr}(X_n^2)\right)^2\right] \longrightarrow 1,$$

proved as follows. One writes

$$\frac{1}{n}\operatorname{Tr}(X_n^2) = \frac{1}{n^2}\sum_{1\le i\le j\le n} c_{ij}W_{ij}^2, \qquad c_{ij}\in\{1,2\}.$$

The variables W_{ij}^2 are independent over the upper triangle, and the uniform fourth-moment bound gives a uniform bound on their variances. Hence the variance of the weighted upper-triangular sum is $O(n^2)$, so the normalized variance is $O(n^{-2})$. Since the expectation is already known to be 1, the second moment tends to 1.

The former T34 integrability gap is now closed for concrete Wigner inputs. In the namespace `WignerInputConcrete`, the theorem `snMonomialIntegrability` derives the monomial spectral-average integrability and squared integrability assumptions from finite sums and products of concrete entry moments. The abstract concentration theorem remains conditional for a general `WignerInput`, as intended, but the concrete supplier is now compiled.

Question 1. The Lean proof follows the same uniqueness principle as the main text: once a candidate sequence has the correct first two values and satisfies the same order-two recurrence, it is equal to u_n. For $|\alpha| < 2$ the proof uses the real identity

$$\sin((m+2)\theta) = 2\cos\theta\,\sin((m+1)\theta) - \sin(m\theta),$$

rather than complex roots. This is the route now displayed in the main text because it is also the shortest concours-friendly verification.

Question 2. The Lean development forced the Riemann-sum proof to keep track of the omitted endpoint intervals and of the harmless normalising factors $(n+1)/n$ and $(2n+1)/(2n)$. The main text has therefore been written with an explicit modulus-of-continuity estimate and a separate bound for the short interval $[mh, 1]$.

Questions 10–12. The no-Weierstrass constraint is preserved in Lean. The formal proof uses only the explicit polynomials

$$Q_n(X) = (1-X^n)^{2^n}, \qquad P_n(X) = Q_n\left(\frac{1-X}{2}\right),$$

then proves uniform step approximation and polynomial replacement. The scaling near a jump c uses

$$\frac{x-c}{1+|c|}$$

so that the argument remains in $[-1,1]$ for $x\in[-1,1]$. Lean also checks the bounds $0\le P_n\le 1$ and the “at most one bad jump” estimate.

Question 17. The compact-support convergence proof is mechanised conditionally on the explicit moment and integrability suppliers stated in the theorem signatures. The formal proof separates the deterministic approximation inequality, the polynomial tail bound, the Chebyshev concentration step, and the finite-union argument. The remaining issue is not the Question 17 argument itself, but the probabilistic model needed to derive all suppliers from the entries of X_n.

Current conclusion. Lean has not found a false statement in the main concours correction. It has, however, made the final probabilistic dependency precise: the prose proof may assume (H_k) after Question 13 as the problem allows, but a first-principles formalisation of Question 13c needs a stronger Wigner model than the current placeholder structure.